

Under Pressure: RA²R and the Emergence of Uninstructed Reasoning Behaviors in Scaffold-Augmented Language Models

Franko Luci
Ejentum
info@ejentum.com

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Disclosure. The author is the founder and sole developer of Ejentum, which commercially offers the Logic API described in this paper. All benchmark data, generation outputs, judgment scores, system prompts, and evaluation metrics are publicly available at <https://github.com/ejentum/benchmarks> under CC BY 4.0. The experimental protocol, including negative findings, is reported in full.

Abstract

We introduce Reasoning Ability-Augmented Retrieval (RA²R), a paradigm for augmenting large language model agents at inference time by retrieving and injecting structured cognitive operations rather than information. Where Retrieval-Augmented Generation (RAG) retrieves facts and Buffer of Thoughts (BoT) retrieves reasoning templates, RA²R retrieves complete cognitive procedures that include named failure mode declarations, executable reasoning topologies, inline epistemic checkpoints, and structured failure recovery mechanisms. We evaluate RA²R across three independent benchmarks using Claude (Anthropic) as the sole model family: EjBench (180 domain-specific tasks, $n = 536$ judgments), a combined suite of BIG-Bench Hard, CausalBench, and MuSR (70 published academic tasks, $n = 209$ judgments), and ARC-AGI-3 (25-step interactive reasoning, $n = 2$ conditions). On single-turn tasks, RA²R injection improved composite reasoning quality by +10.1 percentage points on custom tasks and +20.8 percentage points on published benchmarks, with self-monitoring scores nearly doubling while correctness remained stable. On the interactive benchmark, both conditions scored 0.0 RHAЕ (neither solved the task), but process-level analysis revealed three uninstructed emergent behaviors: spontaneous transition from natural language to symbolic mathematical notation, progressive improvement in retrieval query quality without instruction, and reversal of the expected reasoning decay pattern from -0.005 to $+0.014$ slope across 25 steps, with a scaffold persistence half-life of 24 steps. We report all negative findings, including correctness decrements under multi-ability injection and an unresolved $1.9\times$ increase in normalized contradictions. All data is publicly available. All experiments use a single model family; cross-model generalization is untested.

1 The Pressure Thesis

Large language models fail. We observe that they fail not primarily because they lack knowledge, but because they apply the wrong cognitive operation to the problem at hand. A model with access to every fact in its training corpus will still reverse the direction of a causal chain, confabulate a timeline that never existed, violate the topology of a spatial arrangement, collapse distinct categories into a single abstraction, or generate a confident answer with zero uncertainty signal.

These failure patterns are well-documented in frontier model evaluations [7, 2]: models that achieve near-perfect scores on knowledge benchmarks fail systematically on tasks requiring novel reasoning. These are not information failures. They are structural reasoning failures.

Retrieval-Augmented Generation [10] addressed one half of the problem: the information gap. When a model does not know a fact, RAG retrieves it. The architecture is well-understood: embed the query, search a document store, inject the retrieved passages into context, generate. RAG has become infrastructure. But RAG cannot help a model that knows all the relevant facts and still reasons incorrectly about them. The model does not need more information. It needs a different cognitive procedure.

This paper introduces Reasoning Ability-Augmented Retrieval (RA²R), a paradigm that retrieves *cognitive operations* rather than information. An RA²R system maintains a library of structured reasoning procedures, each designed to address a specific class of cognitive failure. When an agent encounters a reasoning challenge, the system retrieves the appropriate procedure and injects it into the agent’s context before generation. The retrieved artifact is not a document, not a prompt template, and not a tool. It is a complete cognitive operation: a named failure mode, an executable reasoning topology, inline self-monitoring checkpoints, suppression and amplification signals, and a falsification test.

1.1 Six Dimensions of Structural Failure

We organize reasoning failures into six dimensions, each corresponding to a distinct cognitive failure mechanism observable in production LLM agents:

1. **Causal.** The model treats correlation as causation, reverses causal direction, or stops at surface-level symptom attribution. The mechanism: pattern-matching on co-occurrence maps temporal proximity to causal direction.
2. **Temporal.** The model confabulates timelines, blends temporal signals from different epochs, or violates sequence constraints. The mechanism: training data mixes temporal signals without explicit temporal grounding.
3. **Spatial.** The model violates topological constraints, breaks containment boundaries, or fails to maintain spatial state across steps. The mechanism: no persistent spatial representation; all topology is encoded as linguistic implication.
4. **Simulation.** The model cannot maintain counterfactual state, drifts back toward its training distribution, or conflates hypothetical and actual scenarios. The mechanism: the model’s prior is its training distribution, which exerts gravitational pull on all generation.
5. **Abstraction.** The model collapses distinct categories, over-generalizes, or loses critical specifics during abstraction. The mechanism: embedding representations compress semantic distinctions.
6. **Metacognition.** The model produces fluent, confident outputs with no self-monitoring, no uncertainty signal, and no mechanism for detecting its own errors. The mechanism: the training objective rewards fluency, not epistemic accuracy.

1.2 The Thesis

The core thesis of this paper is an asymmetry observation: *telling a model what not to do is more effective than telling it what to do.*

When a cognitive scaffold is injected with a suppression signal (for example, “do not treat correlation as causation”), we hypothesize that the model’s output space is multiplicatively constrained: entire branches of incorrect reasoning are pruned before generation begins. By contrast, an amplification signal (for example, “find the root cause”) is additive: it makes the desired behavior more probable without eliminating the failure mode. The suppression is a gate. The amplification is a nudge. This asymmetry is a design hypothesis derived from our benchmark observations, not a demonstrated causal mechanism. A controlled ablation (suppression-only vs. amplification-only) has not been conducted and is identified as a priority for future work in Section 6.

This asymmetry produces a testable prediction: a model under suppression pressure will not merely avoid the blocked failure mode. It will find alternative paths that it would not have explored without the constraint. The suppression is the pressure. The model’s response to that pressure is the emergence. The instruction is negative. The intelligence is what the model does instead. The pressure thesis is falsified if suppression produces only avoidance (reduced generation diversity, shorter outputs, increased refusals) without generating novel reasoning strategies. If suppression merely silences the model rather than redirecting it, the thesis fails.

We call this the *pressure thesis*, and it is the organizing claim of this paper. Section 2 defines the RA²R paradigm. Section 3 shows the anatomy of one cognitive operation. Section 4 presents evidence from three benchmarks. Section 5 presents the emergent findings that the pressure thesis predicts: uninstructed behaviors that arose from suppression, not from instruction.

2 The RA²R Paradigm

2.1 From Information Retrieval to Reasoning Retrieval

Figure 1 illustrates the architectural distinction. Table 1 presents the structural comparison between RAG and RA²R.

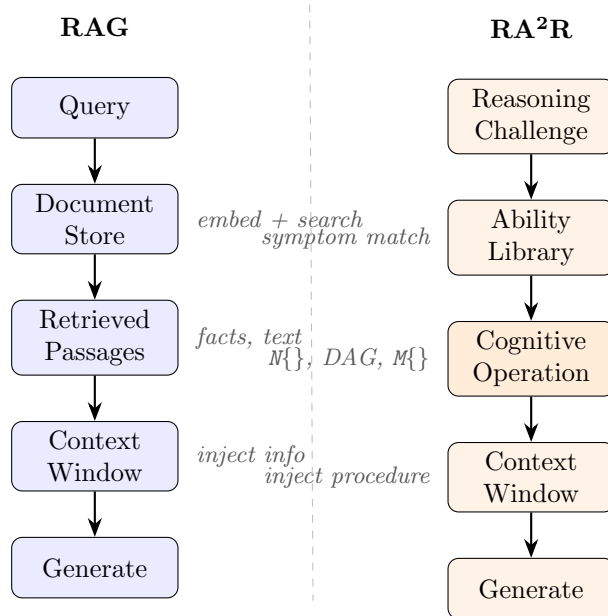


Figure 1: RAG retrieves information to fill knowledge gaps. RA²R retrieves cognitive operations to correct reasoning failures. The retrieved artifact (center row) is the key distinction: passages vs. structured procedures with self-monitoring.

The distinction is not incremental. RAG retrieves facts to fill knowledge gaps. RA²R retrieves cognitive procedures to correct reasoning failures. The retrieval target, the trigger mechanism, the artifact structure, and the post-retrieval behavior are fundamentally different.

Table 1: RAG vs. RA²R: a structural comparison.

Dimension	RAG	RA ² R
What is retrieved	Documents, passages, facts	Cognitive operations (procedures, constraints, self-checks)
Retrieval trigger	Query embedding similarity	Reasoning failure detection (symptom-based matching)
Artifact structure	Unstructured text or passages	23-field typed object: topology, gates, suppression vectors, falsification test
Self-monitoring	None	Inline epistemic checkpoints (M-nodes) with binary pass/fail branching
Failure recovery	None (re-retrieve or fail)	Controlled topology rupture: abandon structured reasoning, freeform corrective analysis, re-enter graph
Composition	Independent documents	Directed synergy graph (requires, enhances, fallback) enabling compound suppression
Delivery timing	During generation	Before generation (pre-injected into reasoning context)

2.2 The Ability Injector

The fundamental unit of RA²R is the *ability injector*: a structured cognitive operation designed to address a specific class of reasoning failure. The current system contains 311 ability injectors distributed across the six cognitive dimensions defined in Section 1 (51–54 abilities per dimension). The complete catalog of abilities, including their names, domains, and trigger conditions, is publicly browsable at <https://ejentum.com/abilities>. Each injector is a typed object with 23 fields. We define the key components:

Negative Gate (N{ })

A first-class declaration of the cognitive failure mode this ability exists to prevent. For example: N{accept_any_causal_assertion_backed_only_by_cooccurrence}. Every ability in the library (311/311) carries an N{ } declaration. This construct has no precedent in the reasoning topology literature; existing frameworks [4, 19, 13] define what a graph *does*, not what it exists to *prevent*. We formalize this as the *anti-pattern trap*: a named failure mode that enables symptom-based retrieval. When an agent’s reasoning exhibits symptoms matching an N{ } declaration, that ability is retrieved.

Reasoning Topology

An executable directed acyclic graph (DAG) specifying the exact reasoning procedure. Topologies are expressed in a domain-specific notation with typed nodes: step nodes (S1:action), gate

nodes ($G1\{\text{condition?}\}$ with $--\text{yes} \rightarrow$ / $--\text{no} \rightarrow$ branches), accumulator nodes ($ACC[\text{items}]$), fork/join nodes, and loop markers. The library contains 24 distinct topology types organized into five families (Table 2). 76.4% of topologies contain iterative loops; the average topology has 6.5 nodes and 2.2 gate decision points.

Table 2: Topology type families (24 types across 5 families). Type names shown; syntax withheld.

Family	Example Types	Count
Control Flow	Gated Pipeline, Fork-Join Compare, Iterative Convergence	6
Structural	Accumulate-Classify-Decide, Convergence Funnel	7
Verification	Bidirectional Verification, Adversarial Self-Attack	5
Reference Frame	Anchor-Drift-Correct, Abstraction Climb-and-Descend	4
Meta-Level	Watchdog Monitor, Emergence Detection	2
		24

Epistemic Gates ($M\{\}$)

Mid-graph self-evaluation checkpoints that force the model to assess the integrity of its own reasoning *before* proceeding to the next step. Each M-node contains a natural-language self-interrogation and binary branching: $--\text{working} \rightarrow$ continues execution; $--\text{failing} \rightarrow$ triggers corrective action. 65.1% of abilities contain at least one M-node (126 total instances across the library). The concept of self-evaluation is not new; Self-Refine [11], Reflexion [14], and CRITIC [8] all evaluate reasoning quality. The architectural distinction is *where* evaluation occurs: these systems operate on completed outputs. M-nodes operate *inside* the reasoning graph, *between* steps, *before* any output exists. The novelty is positional, not conceptual. The following example from a counterfactual reasoning ability illustrates how an M-node operates within a topology:

```
M{Before altering variable B: am I projecting B's
  counterfactual from the frozen baseline, or have I
  unconsciously imported OUTCOME_A's trajectory?
Check: are A and C at their original frozen values?}
--working-> S3b:alter_variable_B
--failing-> ABANDON_GRAPH
  -> FREEFORM{Name the variable that drifted from
    baseline. Reset to frozen state.}
  -> RE-ENTER at S3b
```

Controlled Topology Rupture

The $--\text{failing}$ branch in the example above demonstrates the three-phase recovery mechanism. When an M-node detects reasoning failure, the topology does not simply retry the failed step. It executes: (1) **ABANDON_GRAPH**: exit the structured topology entirely; (2) **FREEFORM**: perform unconstrained natural-language reasoning to diagnose the failure; (3) **RE-ENTER at S_n** : return to the graph at a specific prior node with the corrected understanding. The library contains 81 **FREEFORM** instances with re-entry distances ranging from 1 to 8 nodes back.

Suppression and Amplification Vectors

Each ability carries 1–3 suppression vectors (named failure modes to block) and 2–4 amplification vectors (reasoning patterns to strengthen). Our design hypothesis posits an asymmetry: suppression acts multiplicatively on the output space (pruning branches), while amplification acts additively (increasing probability). These vectors, along with additional cognitive control parameters, are bundled in a structured payload injected before generation.

Falsification Test

A natural-language criterion for detecting whether the ability *failed* to achieve its purpose. For example: “If every causal claim passes without demanding mechanistic evidence, adversarial validation has collapsed.” This functions as a semantic unit test evaluated after execution.

Cognitive Delta

A contrastive pair specifying incorrect versus correct reasoning behavior. For example, the PREDICTIVE MAPPING ability’s cognitive delta contrasts *wrong*: “Skips from start to goal assuming the path is clear without checking intermediate states” with *right*: “Enumerates every intermediate state and verifies feasibility before committing to a transition path.” This bridges topology design (structural) and behavioral specification by embedding expected behavior directly in the ability definition.

Figure 2 illustrates how these constructs compose into a complete reasoning topology. Unlike the simpler PREDICTIVE MAPPING topology (Figure 3), this example includes an epistemic gate (M-node), a controlled topology rupture path, and a convergence loop, demonstrating the full expressiveness of the notation.

2.3 Retrieval Mechanism

Ability retrieval uses embedding-based search matched to detected reasoning failure symptoms. A query describing the agent’s current reasoning challenge is matched against the ability library. Retrieval is deterministic, requires no LLM inference, and completes in under one second.

2.4 Injection Modes

The system supports two injection modes:

Ki (single-ability injection). The single best-matching ability is retrieved and injected. This mode is effective for tasks with a clear, focused reasoning challenge: one failure dimension, one corrective operation. On external academic tasks (Section 4.2), Ki produced a +20.8 percentage point composite improvement.

Haki (multi-ability injection). Three to four abilities with complementary roles are retrieved and composed into a compound scaffold. Selection respects a directed synergy graph that encodes inter-ability dependencies and compatibility constraints. Suppression vectors from all selected abilities are merged, producing *compound suppression*: multiple failure modes blocked simultaneously. On complex multi-variable tasks (Section 4.1), Haki produced a +10.1 percentage point improvement, outperforming Ki’s +9.0 points on the same benchmark.

2.5 The Asymmetry Principle

The design of the ability library is governed by an empirical asymmetry: suppression signals produce larger and more consistent improvements than amplification signals. A suppression vector such as **Suppress: correlation_to_causation_leap** eliminates an entire failure class from the model’s output space. An amplification vector such as **Amplify: root_cause_isolation** increases the probability of correct reasoning without eliminating the failure mode. In both benchmarks (Sections 4.1 and 4.2), the largest per-factor improvements occurred in dimensions most directly addressed by suppression: self-monitoring (+92% on EjBench, +134% on external tasks) and verification (+45%, +84%), both of which are defined by what the model *stops failing to do* rather than what it newly learns to do.

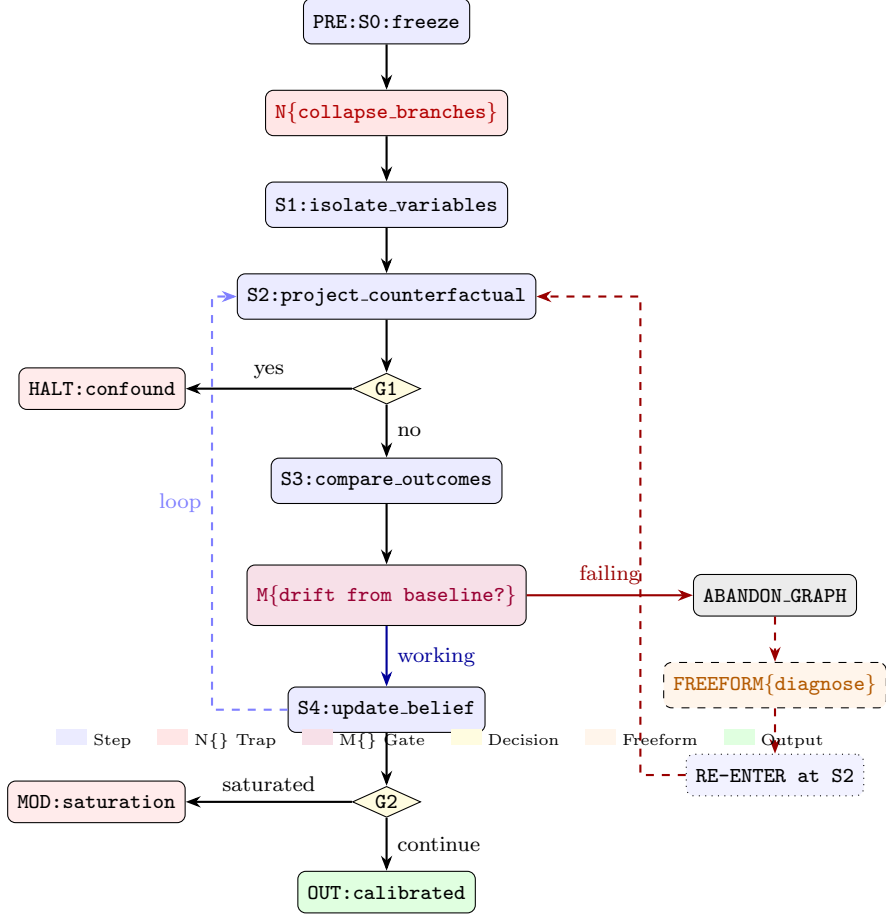


Figure 2: A complete RA²R reasoning topology demonstrating all novel constructs. The main chain flows top-to-bottom through a semantic freeze, anti-pattern trap (red), step nodes (blue), and gate decisions (yellow). The purple M-node forces mid-graph self-evaluation: the `--working` branch (blue arrow) continues execution; the `--failing` branch (red dashed arrows) triggers controlled topology rupture through ABANDON_GRAPH, unconstrained FREEFORM reasoning, and RE-ENTER back at S2. A convergence loop (blue dashed) iterates until saturation is detected.

2.6 Comparison with Buffer of Thoughts

Table 3 compares RA²R ability injectors with the thought-templates introduced by Buffer of Thoughts [17]. BoT established the foundational concept that reasoning strategies can be retrieved from a library. RA²R extends this in nine dimensions.

2.7 Dataset Engineering

The 311 abilities were hand-engineered and subjected to a quality standard we call the *3-Litmus Test*. Every ability must satisfy three criteria simultaneously: (1) it describes a *cognitive operation*, not domain knowledge (“apply Bayesian update to prior beliefs” passes; “compute a discounted cash flow valuation” fails); (2) it is *LLM-executable* using only information available in the agent’s context, requiring no external API calls or database queries; (3) it is *domain-agnostic*, applicable across subject areas without modification. During a purification pass, 96 of 311 abilities (31%) were rewritten to meet this standard, eliminating abilities that appeared cognitively rigorous but

Table 3: BoT thought-template vs. RA²R ability injector.

Feature	BoT Thought-Template	RA ² R Ability Injector
Reasoning procedure	Generic skeleton	Executable DAG with typed nodes, gates, loops
Failure mode declaration	None	N{} anti-pattern trap (100% coverage)
Self-monitoring	None	M{} epistemic gates (65.1% coverage)
Failure recovery	None	ABANDON → FREEFORM → RE-ENTER
Post-execution verification	None	checkpoint field (semantic falsification)
Behavioral specification	None	cognitive_delta (wrong/right pairs)
Cognitive loading	None	Suppress/amplify vectors, style, elasticity
Composition support	Single template	Synergy graph (requires, enhances, fallback)
Retrieval trigger	Embedding similarity	Symptom detection (trigger + keywords)
<i>Where BoT has advantages:</i>		
Computational overhead	Minimal	4.2× token increase (longer traces)
Model independence	Tested across model families	Claude-only (untested on others)

encoded domain knowledge rather than reasoning procedures.

The reasoning topologies across the library exhibit a distribution of edge types that aligns with independently-derived empirical findings. Using the bond-type taxonomy from the molecular structure analysis of long chain-of-thought reasoning [6], which identifies optimal ratios of deductive (60–70%), reflective (20–30%), and exploratory (5–10%) bonds, the RA²R topology library measures at 61.8% deductive, 26.8% reflective, and 11.4% exploratory. This alignment was not an explicit design target.

3 The Scaffold: Anatomy of a Cognitive Operation

To make the ability injector concrete, we present a complete scaffold retrieved during the ARC-AGI-3 interactive reasoning study (Section 4.3). This specific scaffold, PREDICTIVE MAPPING, was retrieved in response to the agent’s query: *“How do I navigate from the robot at (16, 32) through corridor obstacles to reach the goal at (19–54, 61–62) when rows 50–52 appear to be solid walls blocking the direct path south.”*

Listing 1: Complete RA²R scaffold: PREDICTIVE MAPPING. Retrieved in response to a spatial navigation query during the ARC-AGI-3 study.

```
[REASONING INJECTION]
Phase: PREDICTIVE_MAPPING | Confidence: CALIBRATE | Trigger: MOTION_PLAN

[NEGATIVE GATE]
The robot moves from start to end, but intermediate states
are ignored, leaving path feasibility unchecked.

Step 1: Identify the start state and desired end state.
Step 2: Enumerate all intermediate states the transition must
        traverse and map each to feasibility constraints.
Step 3: Check each intermediate for resource sufficiency and
        constraint violations.
Step 4: If any intermediate is infeasible, trace the blockage
        and identify an alternative route. If all pass, validate
        the path end-to-end.
Step 5: Verify that no transition gap exists between consecutive
        states. If a gap is found, reject the path and demand a
        bridge state.
Never skip intermediate validation by jumping from start to end.

[REASONING TOPOLOGY]
S1:identify_start_and_end
  -> N{skip_intermediate_validation}
  -> S2:enumerate_intermediate_states
  -> HOPS[start->I1->I2->...->end]
  -> for_each_hop -> S3:check_feasibility
  -> G1{infeasible?}
    --yes-> HALT:blockage_and_reroute
    --no-> S4:verify_no_transition_gap
  -> G2{gap?}
    --yes-> HALT:demand_bridge_state
    --no-> OUT:validated_transition_path

Amplify: intermediate_state_enumeration; path_feasibility_checking
Suppress: start_end_only_thinking; transition_gap_tolerance

[FALSIFICATION TEST]
If the output does not include a validated transition path or
identify the specific blockage point, verification failed.
```

3.1 Component Walkthrough

Negative Gate. The scaffold opens with a declaration of the failure mode it exists to prevent: jumping from start to end without validating intermediate states. This is the $N\{\}$ construct. It primes the model to recognize the specific cognitive trap before reasoning begins. The agent does not need to be told what to do; it needs to be told what failure looks like.

Step Procedure. Five concrete, sequenced steps that constitute the reasoning operation. Each step has a clear input-output relationship to its predecessor. The procedure is executable by the model using only information available in its context (the game state).

Reasoning Topology. The DAG formalizes the procedure as an executable graph (Figure 3). Step nodes encode reasoning operations, gate nodes enforce binary decisions, and the anti-pattern

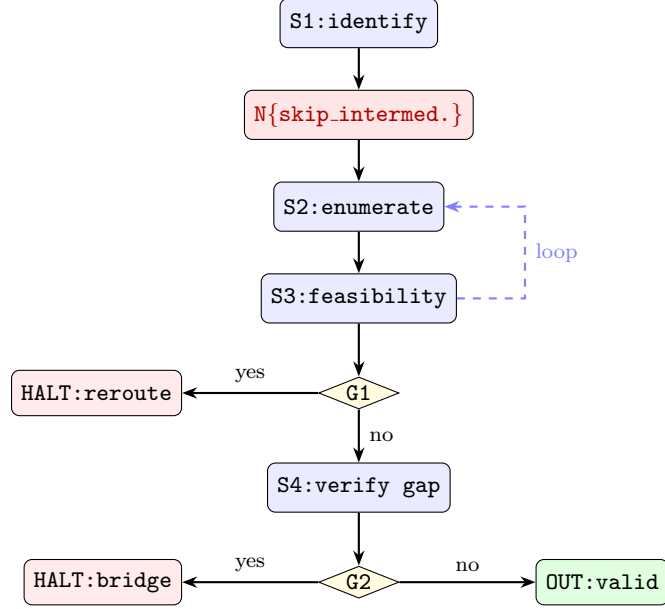


Figure 3: The PREDICTIVE MAPPING reasoning topology as a directed graph. Red: anti-pattern trap ($N\{\}$). Yellow: binary gate nodes. Green: terminal output. Dashed blue: iteration loop over intermediate states.

trap anchors the topology’s purpose. The topology is not a flowchart for the reader; it is an execution specification for the model.

Suppress/Amplify Vectors. Two suppression signals: `start_end_only_thinking` (the core failure mode) and `transition_gap_tolerance` (a secondary failure of accepting gaps between states). Two amplification signals: `intermediate_state_enumeration` and `path_feasibility_checking`. The suppression signals are the operative components; the amplification signals provide positive reinforcement.

Falsification Test. A post-execution check: did the output contain either a validated path or a specific blockage identification? If neither is present, the scaffold failed. This is the semantic unit test.

This scaffold was one of 25 retrieved during the ARC-AGI-3 study. Its suppression signal, `start_end_only_thinking`, directly contributed to one of the emergent behaviors reported in Section 5: a spontaneous shift from natural language to symbolic mathematical notation at step 15 of the interactive session.

4 Experimental Evidence

We evaluate RA²R across three independent benchmarks. All experiments use a single model family (Claude, Anthropic): Opus 4.6 for single-turn tasks and Sonnet 4.6 for the interactive benchmark. This is a limitation; generalization to other model families is untested. Evaluation of single-turn tasks uses an LLM-as-judge protocol; this is a limitation addressed in Section 6.

In all experiments, agents call the RA²R endpoint as a tool (function call) and receive the scaffold in the tool response, which is then included in the agent’s context for the subsequent generation. All data, including generation outputs, judgment scores, system prompts, and raw metrics, is publicly available at <https://github.com/ejentum/benchmarks> under CC BY 4.0.

4.1 EjBench: 180 Domain-Specific Tasks

4.1.1 Design

EjBench contains 180 tasks spanning six cognitive domains (30 per domain: causal, temporal, spatial, simulation, abstraction, metacognitive). Tasks were designed to probe structural reasoning failures rather than factual knowledge, with 54% containing counter-intuitive elements that penalize surface-level reasoning. Formats include multiple-choice (20%), binary yes/no (20%), and open-ended free-text (60%).

4.1.2 Protocol

Three conditions were evaluated: (A) baseline Claude Opus 4.6 with no injection, (B1) single-ability Ki injection, and (C1) multi-ability Haki injection. Agents called the RA²R API as a tool during execution; the system did not pre-select abilities. Evaluation used a two-stage blind protocol: the generator was blind to ground truth and condition labels; the judge (Claude Opus 4.6) was blind to which condition produced each output. A 7-factor rubric scored each generation on a 0–3 scale: correctness, reasoning depth, self-monitoring, verification, epistemic honesty, alternative consideration, and audit trail. The composite quality score is the sum of all factors divided by 21 (range 0–1). Of 540 generations, 536 received valid judgments.

4.1.3 Results

Table 4: EjBench composite results ($n = 536$ judgments, $180 \text{ tasks} \times 3 \text{ conditions}$).

Condition	Composite	Δ from A	n
A (Baseline)	0.621	—	179
B1 (Ki, single)	0.711	+0.090 (+9.0pp)	180
C1 (Haki, multi)	0.722	+0.101 (+10.1pp)	177

Table 5 shows the per-factor breakdown. The largest improvements occur in factors defined by what the model *stops failing to do*: self-monitoring (+0.864 under Haki, +91.5% relative improvement), verification (+0.667, +44.6%), and alternative consideration (+0.473, +34.4%). Correctness remains statistically flat: -0.031 under Ki and -0.112 under Haki. RA²R changes how the model reasons, not what it concludes.

The factor-level pattern has a structural explanation. Self-monitoring, verification, and alternative consideration have low baselines (0.94, 1.50, 1.37 out of 3.0), leaving substantial room for improvement. Audit trail and reasoning depth have high baselines (2.64, 2.44), creating a ceiling effect that limits observable gains. Additionally, the suppression signals in the ability library directly target self-monitoring and verification behaviors (e.g., “do not accept claims without evidence,” “do not commit to a single hypothesis without testing alternatives”), creating a mechanistic link between the intervention and the factors that improve most. An alternative explanation is that the system simply does not improve depth and audit trail because the suppression signals do not target those behaviors. We cannot fully distinguish these explanations without per-factor ablation.

Negative findings. Haki injection produced a correctness decrement of -0.112 on the 3-point scale. On the spatial domain specifically, Haki composite *decreased* by -2.8 percentage points (from 0.619 to 0.591), the only domain-condition pair showing regression.

Cross-domain retrieval and domain-agnosticity. Domain-level retrieval precision was 38.33%: 62% of tasks received an ability from a mismatched cognitive domain. This initially

Table 5: EjBench per-factor results (0–3 scale).

Factor	A	B1	B1 Δ	C1	C1 Δ
Self-Monitoring	0.944	1.700	+0.756	1.808	+0.864
Verification	1.497	2.011	+0.514	2.164	+0.667
Alt. Consideration	1.374	1.772	+0.398	1.848	+0.473
Epistemic Honesty	1.542	1.900	+0.358	1.944	+0.402
Reasoning Depth	2.436	2.539	+0.103	2.571	+0.135
Audit Trail	2.643	2.750	+0.108	2.763	+0.120
Correctness	2.603	2.572	−0.031	2.492	−0.112

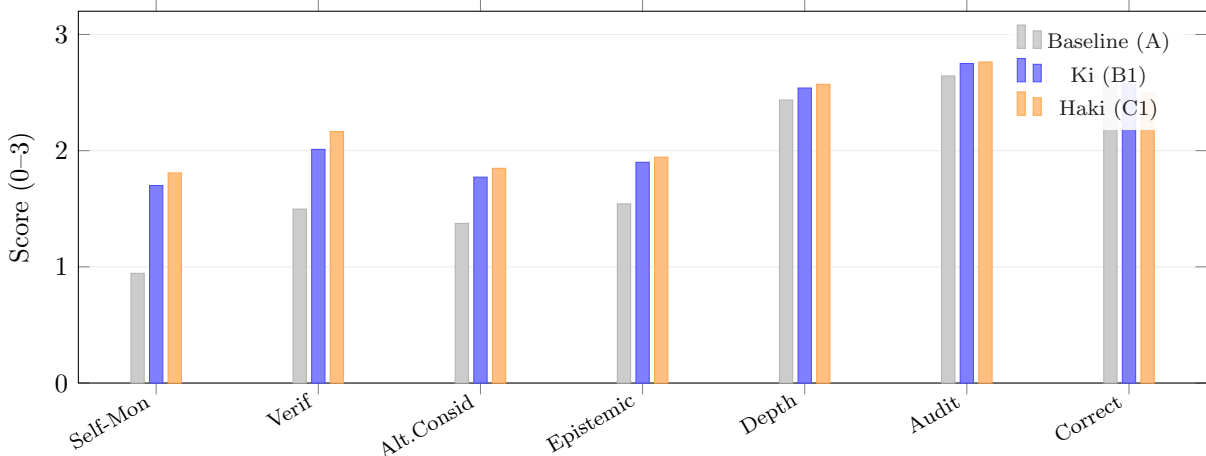


Figure 4: EjBench per-factor scores (0–3 scale). Factors ordered by improvement magnitude. RA²R injection lifts every factor except correctness, which remains flat (gray baseline) or slightly decreases. The largest gains occur in self-monitoring (+91%), verification (+45%), and alternative consideration (+34%).

appears to undermine the retrieval mechanism. However, composite improvements persisted across all mismatched conditions. We interpret this as evidence that the suppression signals operate at the cognitive-procedure level rather than the domain level. This is by design: every ability in the library satisfies the 3-Litmus Test (Section 2), which requires domain-agnosticity. A suppression signal such as “do not accept claims without mechanistic evidence” constrains reasoning regardless of whether the task is causal, temporal, or spatial. This finding is directly relevant to falsification condition F2 (Section 6): if random scaffolds produced equivalent lift, the specific ability would not matter. The fact that *mismatched* abilities still improve quality, while *matched* abilities improve it more, suggests that the suppression mechanism is robust but not arbitrary.

4.2 BBH, CausalBench, MuSR: 70 Published Academic Tasks

4.2.1 Design

To validate on independently-designed tasks, we evaluated on 70 published problems from three established benchmarks: BIG-Bench Hard (25 tasks: 10 causal, 10 temporal, 5 spatial navigation), CausalBench (30 causal reasoning tasks involving DAGs, do-calculus, and counterfactuals), and MuSR (15 theory-of-mind and spatial state tracking tasks). The protocol, rubric, and conditions were identical to EjBench.

4.2.2 Results

Table 6: BBH/CausalBench/MuSR composite results ($n = 209$ judgments).

Condition	Composite	Δ from A	n
A (Baseline)	0.476	—	69
B1 (Ki, single)	0.684	+0.208 (+20.8pp)	70
C1 (Haki, multi)	0.562	+0.086 (+8.6pp)	70

The +20.8 percentage point improvement under Ki on external tasks is the largest single-benchmark gain observed. Correctness improved by +0.14 on the 3-point scale, replicated across two independent runs. Self-monitoring more than doubled, from 0.74 to 1.73 (+134%).

Mode-task alignment. A directional reversal is observed between the two benchmarks: Ki outperforms Haki on the focused, single-answer external tasks (+20.8pp vs. +8.6pp), while Haki outperforms Ki on the complex, multi-variable EjBench tasks (+10.1pp vs. +9.0pp). This is consistent with the compound suppression mechanism: single-ability injection is sharper when one failure dimension dominates; multi-ability injection provides broader coverage when multiple failure dimensions interact.

Negative findings. Haki correctness decreased by -0.12 . On the five spatial navigation tasks from BIG-Bench Hard, Haki composite dropped to 0.238 (from 0.438 baseline), a severe regression of -20.0 percentage points. This replicates the EjBench spatial regression under multi-ability injection and suggests that compound suppression interferes with spatial constraint tracking. Spatial reasoning may require specialized handling under multi-ability injection where multiple suppression signals compete for attention resources that would otherwise maintain spatial state.

4.3 ARC-AGI-3: Interactive Multi-Step Reasoning

4.3.1 Design

ARC-AGI-3 [2] is an interactive reasoning benchmark designed to test general intelligence through novel puzzle-solving. It is unbeaten by frontier models, resistant to memorization (procedurally generated), and scores on action efficiency against a human baseline. We selected game LS20 (keyboard-controlled spatial navigation, 7 levels) and ran 25 steps per condition with Claude Sonnet 4.6. Condition A received the official ARC-AGI-3 system prompt. Condition B received the same prompt plus a 1,354-character protocol layer that instructs the agent on how to request and absorb reasoning scaffolds (but contains no game strategy or domain information). The agent in condition B chose its own retrieval queries and injection modes at each step.

4.3.2 Results

Outcome. Both conditions scored $\text{RHA}E = 0.0$ (neither completed any of the 7 levels). This is consistent with the benchmark’s $<1\%$ frontier solve rate and is reported without qualification.

Process metrics. Despite identical outcomes, the reasoning process diverged substantially. Table 7 presents the key process-level metrics.

Cost. The augmented condition consumed $4.2\times$ more tokens (\$8.48 vs. \$2.88 at published API rates). This cost is driven by longer reasoning traces (mean 731 characters per step vs. 229), not retrieval overhead (which adds <1 second per step). For tasks where reasoning quality, auditability, and self-monitoring are more valuable than throughput, this tradeoff is favorable.

Table 7: ARC-AGI-3 process metrics (LS20, 25 steps per condition, $n = 1$ per condition).

Metric	Baseline (A)	Augmented (B)	Δ
RHAE	0.0	0.0	0.0
Stuck episodes	2	1	-1
Memory decay slope	-0.005	+0.014	+0.019
Scaffold echo rate	0.0	1.12/step	+1.12
Scaffold half-life	0	24 steps	+24
Reasoning depth trend	+0.86	+10.50	$\times 12.2$
Vocab. diversity trend	-0.079	+0.415	reversal
Consistency score	0.16	0.24	+0.08
Contradiction rate	0.28	2.24	$\times 8.0$
Total tokens	84,521	356,768	$\times 4.2$

Limitation. These results represent a single run ($n = 1$) per condition. No statistical significance can be claimed. The metrics are reported as observations that motivate the emergent findings in Section 5, not as demonstrated effects.

5 Under Pressure: Emergent Findings

This section presents five behavioral observations from the ARC-AGI-3 interactive study that were not designed, not instructed, and not predicted by the scaffold content. These observations are consistent with emergence from the interaction between suppression pressure and the model’s own optimization dynamics, but we do not claim demonstrated emergence. Each observation is reported at its actual evidence strength ($n = 1$ per condition for the interactive study), and no observation is presented as a generalizable finding. We report them because they are visible in the data, because they are directionally consistent with the pressure thesis, and because they define a concrete research program for future work with larger sample sizes.

5.1 Spontaneous Domain Shift

At step 15 of the augmented condition, the agent spontaneously transitioned from natural language spatial reasoning to symbolic mathematical notation. The transition is visible in the raw reasoning traces.

Baseline, step 15 (377 characters):

“Player (#) is at col 16, row 32. The goal (!) is at rows 61–62, cols ~40–57. Need to navigate east and south. There’s a wall at cols 25–29 for rows 30–39, but the wide corridor at rows 25–29 spans cols 10–53 allowing eastward travel.”

Augmented, step 15 (1,000 characters, truncated at measurement limit):

“Negative gate acknowledged: not reasoning purely in natural language without symbolic analysis. S1: Define symbols. Robot R=(rx,ry). Let rx~20, ry~32 (reading row 32 col: 4 spaces + 10 stars + 6 pluses = offset 20, ‘#’ at x=20). Goal G=(gx,gy) where...”

The cause is identifiable. The scaffold’s **Suppress: start_end_only_thinking** signal constrained the agent’s reasoning modality. It did not instruct the agent to use mathematical notation; it blocked the agent from reasoning only about start and end states. Facing this constraint, the agent independently adopted symbolic variable assignment as a means of tracking intermediate

states. The scaffold applied pressure (suppression of a failure mode). The model’s response to that pressure (symbolic notation) was its own adaptation.

This observation is $n = 1$ (single step, single game). We do not claim generality. We note that the transition was not prompted by the scaffold content, which contains no mathematical notation, no variable assignment syntax, and no instruction to change reasoning modality.

5.2 Query Quality Evolution

The augmented agent was required to provide a natural-language query for each RA²R retrieval call, describing its current reasoning challenge. Over 25 steps, the queries evolved through five qualitative stages without any instruction on query formatting or specificity:

1. **Generic** (step 1): “I’m analyzing a grid-based game state to determine which of 4 actions to take.”
2. **Spatially specific** (step 7): “How do I navigate from (16,32) through corridor obstacles when rows 50–52 appear to be solid walls.”
3. **Metacognitive** (step 10): “Robot appears stuck at (16,32) after 5 turns trying actions 2,3,4.”
4. **Precise** (step 17): “Wall barrier at x=29–33 for rows 30–39 blocks rightward movement.”
5. **Solution-oriented** (step 22): “How to navigate around the wall at x=29 using the corridor at y=25–29.”

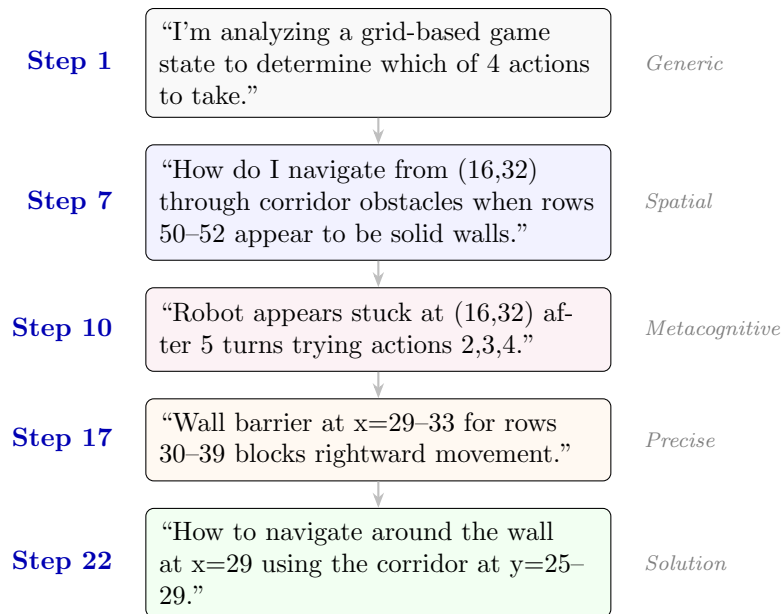


Figure 5: Query quality evolution over 25 steps. The agent was told *that* it must provide a query, not *how* to formulate one. Queries evolved from generic game descriptions to solution-oriented spatial reasoning with precise coordinates. No instruction on query formatting was provided.

The agent learned to be a better user of the retrieval system. Early queries retrieved generic spatial abilities; later queries retrieved precisely-targeted path validation procedures. This is an emergent tool-use skill: the agent was told *that* it must provide a query, not *how* to construct an

effective one. The evolution from “analyzing a grid” to “navigate around the wall at $x=29$ using the corridor at $y=25-29$ ” represents spontaneous optimization of the human-tool interface.

5.3 Memory Decay Reversal

In the baseline condition, reasoning quality degrades over time. The memory decay slope, computed as a linear regression over step-by-step reasoning metrics (back-references, spatial precision, vocabulary diversity), is -0.005 per step: a slow but consistent decline. This is consistent with the known “lost in the middle” effect in long-context language model inference.

In the augmented condition, the slope reverses to $+0.014$ per step: reasoning quality *improves* over time. Scaffold-derived language (terms such as “negative gate,” “intermediate validation,” “suppress”) appeared at a rate of 1.12 instances per step and persisted with a half-life of 24 steps, covering the entire 25-step game session. The compounding slope was $+0.007$, meaning each step’s reasoning built incrementally on the scaffold effects of prior steps rather than diluting them.

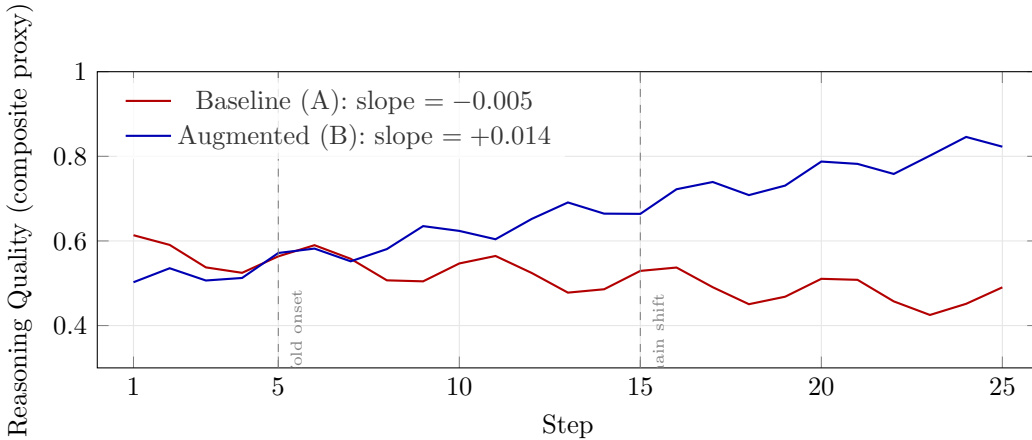


Figure 6: Memory decay reversal. Baseline reasoning quality degrades over 25 steps (red, slope -0.005). Augmented reasoning quality improves (blue, slope $+0.014$). Scaffold-derived language persists with a half-life of 24 steps. Vertical markers: scaffold onset (step 5, first explicit citation) and spontaneous domain shift (step 15).

The mechanism behind this reversal is an open question. One hypothesis is that the structurally distinctive scaffold notation (the DAG format with its symbolic tokens) resists the attention dilution that degrades reasoning in long contexts [16]. This is consistent with research showing that structurally salient tokens receive disproportionate attention weight in transformer architectures. If this hypothesis is correct, the scaffold’s value should increase monotonically with task length. We have not tested this prediction directly; it is stated as a falsifiable hypothesis for future work.

5.4 Cross-Game Validation

The scaffold absorption pattern observed in LS20 was replicated in a separate pilot study on FT09, a different ARC-AGI-3 game using click-based pattern matching (a different input modality from LS20’s keyboard navigation). Despite the change in game mechanics, input modality, and visual structure, the augmented agent produced the same behavioral markers:

- Explicit scaffold citations (“Applying scaffold: extracted landmarks”)

- Named suppression signals (“Suppressed all_points_equal bias”)
- Scaffold-structured spatial analysis

The patterns replicate across two games (LS20, FT09), three dates, and four step budgets (3, 5, 25, 33 steps). This is consistent with the scaffold’s effect being driven by its structural properties rather than game-specific content.

5.5 The Contradiction Increase

The augmented condition produced $8\times$ more explicit contradictions than the baseline (56 vs. 7 across 25 steps). Token-normalized, the ratio is $1.9\times$ (the augmented condition also produced $3.2\times$ more total reasoning tokens). This is a negative finding that we report without resolution.

Two interpretations are possible. First, the contradictions may represent *destructive interference*: multiple suppression signals from the compound scaffold creating conflicting constraints that the model cannot simultaneously satisfy. Second, the contradictions may represent *productive cognitive conflict*: the model explicitly naming points where its observations contradict its prior beliefs, a behavior that precedes belief revision. The scaffold’s **NEGATIVE GATE** and **FALSIFICATION TEST** components mechanically increase contradiction-adjacent language (terms like “however,” “but this contradicts,” “this is inconsistent with”), which inflates the raw metric.

We lack sufficient data to distinguish between these interpretations. The contradiction increase is a real, measured phenomenon. Its valence (beneficial or harmful) is an open question.

6 What Falsifies This

The core claim of this paper is that structured reasoning injection produces measurable behavioral changes in LLM outputs, and that suppression-driven injection produces uninstructed emergent behaviors. We identify four conditions under which these claims are falsified:

F1: Zero behavioral change on a different model family. If RA²R injection produces no measurable improvement on GPT-4, Gemini, or Llama models, the effect is model-specific rather than a property of transformer-based language models under scaffolding. *Status: untested.* We expect generalization (the mechanism operates at the attention level, which is shared across transformer architectures), but this is a hypothesis, not a finding.

F2: Random scaffold injection produces equivalent lift. If shuffled suppression signals, mismatched topologies, or randomly-generated scaffolds produce the same composite improvement as semantically-matched abilities, the specific cognitive operation does not matter and the effect is attributable to prompt length, structured formatting, or instruction-following compliance. *Status: untested.* This is the most critical ablation and a priority for future work.

F3: Evaluator bias systematically inflates injected conditions. If the LLM judge assigns higher scores to longer, more structured outputs regardless of reasoning quality, the improvements reflect evaluation bias rather than genuine behavioral change. *Status: partially addressed.* The two-stage blind protocol prevents the judge from knowing which condition produced each output. However, the judge may learn to associate structural markers (scaffold-derived language) with higher quality even without explicit condition labels. Human evaluation is needed to fully address this condition.

F4: Replication produces directionally different results. If an independent replication of the benchmark protocol produces improvements in different factors, degradation where we report improvement, or no composite change, our results may reflect noise or idiosyncratic prompt

interactions. *Status: partially addressed.* The correctness improvement on external tasks (+0.14) was replicated across two independent runs with consistent direction. The full 7-factor profile has not been replicated.

7 Related Work

7.1 Reasoning Topology Frameworks

The formalization of LLM reasoning as graph-structured computation has progressed from chains [19] to trees to arbitrary graphs [4]. Besta et al. [5] established the taxonomy, defining topology as the algorithm specification rather than a diagram. Graph of Thoughts [4] introduced aggregation and refinement loops, achieving 62% improvement over Tree of Thoughts on sorting tasks. Network of Thought [9] formalized typed nodes (fact, subgoal, constraint, conclusion) and typed edges guided by heuristic controllers. Adaptive Graph of Thoughts [13] constructs DAGs dynamically at test time, expanding only sub-problems judged sufficiently complex.

RA²R extends this progression in a specific direction: making reasoning topologies *retrievable and injectable*. Rather than constructing a topology per-task, RA²R maintains a pre-designed library of topologies matched to cognitive failure classes and retrieves them based on detected symptoms.

7.2 Reasoning Strategy Retrieval

Buffer of Thoughts [17] established the foundational concept: reasoning strategies can be stored in a meta-buffer and retrieved via embedding similarity. BoT reported significant improvements on Game of 24 (51% to 82%), geometric shapes, and Checkmate-in-One tasks, demonstrating that reasoning strategy retrieval is a viable paradigm. BoT’s “thought-templates” are generic reasoning skeletons (decompose, solve sub-problems, recombine). ReasonFlux [18] extends BoT with hierarchical reinforcement learning to select sequences of templates.

RA²R builds on BoT’s foundational insight but differs in the richness of the retrieved artifact. Table 3 details the comparison across nine dimensions. The core distinction: BoT retrieves a skeleton; RA²R retrieves a complete cognitive operation that includes what the skeleton prevents ($N\{\}$), how the skeleton monitors itself ($M\{\}$), and what happens when the skeleton fails (controlled topology rupture).

7.3 The Molecular Structure of Reasoning

Recent work from ByteDance [6] analyzed long chain-of-thought reasoning through a molecular-bond framework, identifying three bond types: deep-reasoning (covalent, 60–70%), self-reflection (hydrogen, 20–30%), and self-exploration (van der Waals, 5–10%). The RA²R topology library’s edge distribution (61.8% / 26.8% / 11.4%) aligns with these empirically-derived optimal ratios without having targeted them. The controlled topology rupture mechanism introduces a behavior that the molecular framework cannot classify: a temporary exit from structured reasoning into unconstrained reflection, followed by controlled re-entry at a prior graph node. This is distinct from covalent, hydrogen, and van der Waals bonds.

Topological data analysis approaches [12] have shown that strong reasoning models exhibit greater cyclicity, broader exploration diameter, and small-world characteristics. RA²R topologies exhibit all three properties: 76.4% contain loops (cyclicity), re-entry distances reach 8 nodes

(diameter), and semantic freeze anchors (`PRE:S0:freeze`) provide shortcut access to grounding definitions (small-world connectivity).

7.4 Self-Correction and Reflection

Self-Refine [11] introduced iterative self-critique at the output level. Reflexion [14] stores verbal reflections on past failures for subsequent attempts. CRITIC [8] adds tool-interactive verification of completed outputs. All three systems operate on finished outputs.

RA²R’s M-node mechanism operates at a different intervention point: inside the reasoning graph, between steps, before any output is generated. This is analogous to the difference between post-mortem analysis and an immune system: M-nodes detect and respond to reasoning failures as they occur, rather than evaluating the final result.

7.5 Retrieval-Augmented Reasoning

RARE [15] adds retrieval actions to MCTS-based reasoning, retrieving *factual documents* during reasoning steps. Self-RAG [3] learns when retrieval is helpful. Both systems retrieve information to support reasoning accuracy. RA²R retrieves *cognitive operations* to augment reasoning procedure. The retrieval target is fundamentally different: facts vs. procedures.

7.6 Training-Time Approaches

Constitutional AI and RLHF embed behavioral constraints into model weights during training. The Meta-Abilities Alignment framework [1] explicitly decomposes cognitive skills (deduction, induction, abduction) and aligns models through RL-generated self-verifiable tasks. RA²R shares the decomposition insight but delivers it at inference time rather than training time. This makes the intervention reversible (scaffolds are injected per-task and removed afterward), composable (multiple abilities can be combined dynamically), and auditable (the exact scaffold injected at each step is logged).

8 Conclusion

This paper contributes four things:

A paradigm. RA²R defines a new category of retrieval: not information (RAG), not reasoning templates (BoT), but complete cognitive operations with self-monitoring, failure recovery, and behavioral specification. The retrieved artifact is richer than any prior work in this space.

A mechanism hypothesis. The observed asymmetry between suppression and amplification. Our data is consistent with suppression acting multiplicatively (pruning failure branches) and amplification acting additively (nudging toward correct reasoning). Across 250 tasks and two benchmarks, the largest improvements occurred in dimensions defined by what the model stopped failing to do. This asymmetry has not been isolated via ablation and remains a design hypothesis.

Evidence. Three independent benchmarks totaling 250+ tasks and 745+ blind judgments. Composite quality improved by +10.1 percentage points on custom tasks and +20.8 percentage points on published academic tasks. Correctness remained stable. Self-monitoring nearly doubled. All data is public.

Observations. Five uninstructed emergent behaviors in an interactive reasoning setting: spontaneous domain shift from natural language to symbolic mathematics, progressive improvement in retrieval query quality, reversal of the expected memory decay pattern, cross-game replication of

scaffold absorption, and an unresolved increase in explicit contradictions. These observations are small- n and do not constitute demonstrated effects. They constitute a thesis: suppression is pressure, and emergence is the model’s response to that pressure.

8.1 What Remains

Five open questions require future work: (1) cross-model validation on GPT, Gemini, and open-source families; (2) ablation of suppression vs. amplification to isolate the asymmetry mechanism; (3) human evaluation to address LLM-as-judge limitations; (4) the random-scaffold control (F2 in Section 6), which is the most critical missing experiment; and (5) replication at scale across more tasks, more runs, and more evaluators.

8.2 Closing

The instruction is what not to do. The intelligence is what the model does instead. The scaffold does not contain symbolic mathematics; the agent invented it. The scaffold does not instruct query optimization; the agent learned it. The scaffold does not specify reasoning persistence; the agent’s reasoning improved over time while the unscaffolded agent’s decayed.

We do not claim that RA²R makes models smarter. We claim that it makes their reasoning visible, governable, and auditable, and that under the pressure of well-designed suppression constraints, models produce behaviors that nobody designed. The pressure thesis is not proven. The data is consistent with it, and the data is public.¹

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¹All benchmark data, generation outputs, judgment scores, system prompts, raw traces, and evaluation metrics: <https://github.com/ejentum/benchmarks> (CC BY 4.0).

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A EjBench Per-Domain Factor Breakdown

Table 8: EjBench results by domain (composite quality score, 0–1 scale).

Domain	A	B1 (Ki)	B1 Δ	C1 (Haki)	C1 Δ
Abstract	0.627	0.753	+0.126	0.820	+0.193
Simulation	0.513	0.677	+0.164	0.699	+0.186
Causal	0.637	0.732	+0.095	0.778	+0.141
Metacognitive	0.743	0.801	+0.058	0.828	+0.085
Spatial	0.619	0.693	+0.074	0.591	−0.028
Temporal	0.587	0.608	+0.021	0.608	+0.021

B BBH/CausalBench/MuSR Per-Source Results

Table 9: External benchmark results by source (composite quality score, 0–1 scale).

Source	Tasks	A	B1 (Ki)	B1 Δ
CausalBench	30	0.498	0.708	+0.210
MuSR	15	0.475	0.698	+0.223
BIG-Bench Hard	25	0.444	0.633	+0.189
Combined	70	0.476	0.684	+0.208

C ARC-AGI-3 Trace Comparison

Table 10: ARC-AGI-3 trace-level metrics (25 steps per condition, totals).

Metric	Baseline (A)	Augmented (B)	Ratio
Total reasoning length (chars)	5,720	18,292	3.2×
Total back-references	4	6	1.5×
Total contradictions	7	56	8.0×
Total scaffold echoes	0	28	—
Total spatial terms	122	172	1.4×
Total unique terms	556	1,310	2.4×

D Additional Figures

Domain	Correct	Depth	Self-Mon	Verif	Epistemic	Alt.C	Audit
Abstract	-0.03	+0.30	+1.30	+1.07	+0.80	+0.70	+0.33
Simulation	0.00	+0.50	+1.27	+1.07	+0.60	+0.83	+0.30
Causal	-0.14	+0.10	+0.93	+0.79	+0.41	+0.66	+0.31
Meta	+0.07	0.00	+0.77	+0.57	+0.33	+0.37	+0.10
Spatial	-0.56	-0.13	+0.48	+0.29	+0.05	+0.05	-0.22
Temporal	-0.03	+0.01	+0.41	+0.20	+0.19	+0.21	-0.11

Figure 7: EjBench domain $\times$ factor heatmap (Haki/C1 delta from baseline, 0–3 scale). Green: improvement. Red: regression. The spatial domain shows correctness regression (-0.56), the only severe negative. Self-monitoring improves across all six domains.

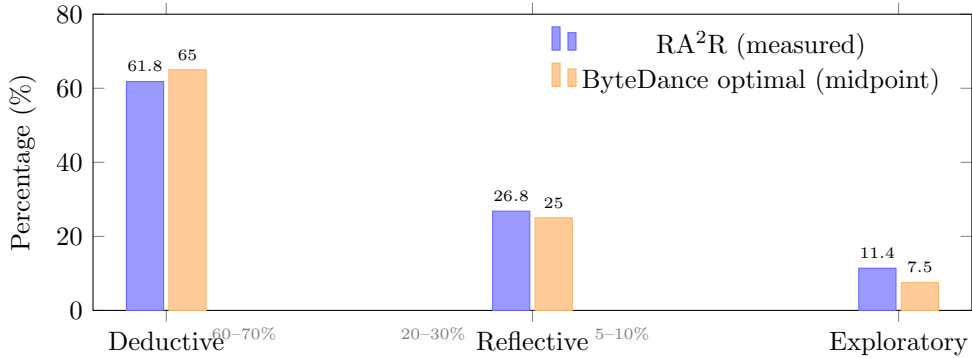


Figure 8: RA²R topology bond distribution vs. ByteDance empirically-derived optimal ratios. The RA²R library (blue) falls within the optimal range for all three bond types without having explicitly targeted these ratios. Gray ranges show the ByteDance empirical optimum.

Step	Baseline (A)	Augmented (B)
1	<i>“Looking at the game state, I can see a grid with various symbols.”</i> 66 ch	“I’m analyzing a grid-based game state to determine which of 4 actions to take. Need to understand map structure.” 699 ch
10	<i>(minimal output)</i> 7 ch	“Robot appears stuck at (16,32) after 5 turns. Applying scaffold: intermediate validation...” 1000 ch
15	<i>“Player (#) is at col 16, row 32. The goal (!) is at rows 61–62. Need to navigate east and south.”</i> 377 ch	“Negative gate acknowledged. S1: Define symbols. Robot R=(rx,ry). Let rx≈20, ry≈32...” 1000 ch
25	<i>“Moving down to continue exploring the lower section of the map.”</i> 345 ch	“Scaffold: verify transition gaps. The corridor at y=25–29 connects to...” 934 ch

Figure 9: Side-by-side reasoning traces at steps 1, 10, 15, and 25. Baseline traces are short and repetitive. Augmented traces grow in length and sophistication, with scaffold echoes (**bold**) persisting throughout. Step 15 (highlighted) shows the spontaneous shift to symbolic mathematical notation.

E Evaluation Rubric

Table 11 defines the 7-factor rubric used for blind evaluation in EjBench and BBH/CausalBench/MuSR. Each factor is scored 0–3.

Table 11: 7-factor evaluation rubric. Score 0 = absent, 1 = minimal, 2 = adequate, 3 = thorough.

Factor	Definition (score 3 = thorough)
Correctness	Reaches the right answer through valid reasoning. Partial credit for correct method with arithmetic/detail errors.
Reasoning Depth	Traces second- and third-order effects. Explores causal chains beyond the immediate question. Multi-level analysis.
Self-Monitoring	Continuously checks its own reasoning. Questions assumptions mid-chain. Detects and flags potential errors before concluding.
Verification	Tests conclusions via independent checks: counterfactuals, boundary cases, re-derivation from different premises.
Epistemic Honesty	Distinguishes established facts from working assumptions. Calibrates confidence. Flags uncertainty explicitly.
Alt. Consideration	Systematically evaluates competing hypotheses. Eliminates alternatives with evidence rather than ignoring them.
Audit Trail	Labels reasoning steps explicitly. Shows intermediate values. Produces a chain that a reader can trace and verify.

F ARC-AGI-3 Protocol Layer

The following protocol layer (1,354 characters) was appended to the official ARC-AGI-3 system prompt for Condition B (augmented). It instructs the agent on *how to request and absorb* reasoning scaffolds but contains no game strategy, domain knowledge, or action recommendations. Condition A received only the official prompt (layers 1–2). Both conditions used identical JSON response schemas for game actions.

Listing 2: ARC-AGI-3 Condition B protocol layer (Layer 3).

```
## Reasoning Augmentation (Active Every Turn)
Before each action, you will receive a [REASONING SCAFFOLD]
block -- a structured reasoning procedure that prevents
common failure modes. The scaffold augments HOW you reason,
not what you know.

### Your choice each turn
You choose the augmentation mode by setting augment_mode:
- "single": focused reasoning on one domain (spatial,
  causal, etc.)
- "multi": complex cross-domain analysis (e.g., spatial +
  causal + metacognitive)
You must also provide augment_query: a 1-2 sentence
description of your current reasoning challenge for
this turn.

### When you receive the scaffold
You MUST:
1. Read [NEGATIVE GATE] first -- the reasoning trap to avoid
2. Follow [REASONING TOPOLOGY] as your decision structure
3. Apply Suppress: signals as a POST-CHECK on your reasoning
4. Verify against [FALSIFICATION TEST]
Then choose your action.
```